CHAPTER 1

An Introduction to *Cambridge Handbook of Expertise and Expert Performance*: Its Development, Organization, and Content

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A significant milestone is reached when a field of scientific research matures to a point warranting publication of its first handbook. A substantial body of empirical findings, distinctive theoretical concepts and frameworks, and a set of new or adapted methods justify a unifying volume. The growth of this field is evident from the publication of a series of edited books on diverse sets of skills and expertise from many domains during the last several decades (Anderson, 1981; Bloom, 1985a; Chase, 1973; Chi, Glaser, & Farr, 1988; Ericsson, 1996a; Ericsson & Smith, 1991a; Feltovich, Ford, & Hoffman, 1997; Hoffman, 1992; Starkes & Allard, 1993; Starkes & Ericsson, 2003). And as in many other fields, the name of a branch of scientific study, in our case expertise and expert performance, often communicates the domain of studied phenomena.

Expert, Expertise, and Expert Performance: Dictionary Definitions

Encyclopedias describe an ***Expert*** as "one who is very skillful and well-informed in some special field" (*Webster's New World Dictionary*, 1968, p. 168), or "someone widely recognized as a reliable source of knowledge, technique, or skill whose judgment is accorded authority and status by the public or his or her peers. Experts have prolonged or intense experience through practice and education in a particular field" (Wikipedia, 2005). ***Expertise*** then refers to the characteristics, skills, and knowledge that distinguish experts from novices and less experienced people. In some domains there are objective criteria for finding experts, who are consistently able to exhibit superior performance for representative tasks in a domain. For example, chess masters will almost always win chess games against recreational chess players in chess tournaments, medical specialists are far more likely to diagnose a disease correctly than advanced medical students, and professional musicians can perform pieces of music in a manner that is unattainable for less skilled musicians. These types of superior reproducible performances of representative tasks capture the essence of the respective domains, and authors have been encouraged

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to refer to them as ***Expert Performance*** in this handbook.

In some domains it is difficult for nonexperts to identify experts, and consequently researchers rely on peer-nominations by professionals in the same domain. However, people recognized by their peers as experts do not always display superior performance on domain-related tasks. Sometimes they are no better than novices even on tasks that are central to the expertise, such as selecting stocks with superior future value, treatment of psychotherapy patients, and forecasts (Ericsson & Lehmann, 1996). There are several domains where experts disagree and make inconsistent recommendations for action, such as recommending selling versus buying the same stock. For example, expert auditors' assessments have been found to differ more from each other than the assessments of less experienced auditors (Bédard, 1991). Furthermore, experts will sometimes acquire differences from novices and other people as a function of their repetitive routines, which is a consequence of their extended experience rather than a cause for their superior performance. For example, medical doctors' handwriting is less legible than that of other health professionals (Lyons, Payne, McCabe, & Fielder, 1998). Finally, Shanteau (1988) has suggested that "experts" may not need a proven record of performance and can adopt a particular image and project "outward signs of extreme self-confidence" (p. 211) to get clients to listen to them and continue to offer advice after negative outcomes. After all, the experts are nearly always the best qualified to evaluate their own performance and explain the reasons for any deviant outcomes.

When the proposal for this *Handbook* was originally prepared, the outline focused more narrowly on the structure and acquisition of highly superior (expert) performance in many different domains (Ericsson, 1996b, 2004). In response to the requests of the reviewers of that proposal, the final outline of the handbook covered a broader field that included research on the development of expertise and how highly experienced individuals accumulate knowledge in their respective domains and eventually become socially recognized experts and masters. Consequently, to reflect the scope of the *Handbook* it was entitled the *Cambridge Handbook of Expertise and Expert Performance*. The current handbook thus includes a multitude of conceptions of expertise, including perspectives from education, sociology, and computer science, along with the more numerous perspectives from psychology emphasizing basic abilities, knowledge, and acquired skills. In this introductory chapter, I will briefly introduce some general issues and describe the structure and content of the *Handbook* as it was approved by Cambridge University Press.

Tracing the Development of Our Knowledge of Expertise and Expert Performance

Since the beginning of Western civilization there has been a particular interest in the superior knowledge that experts have in their domain of expertise. The body of knowledge associated with the domain of expertise in which a person is expert is a particularly important difference between experts and other individuals. Much of this knowledge can be verbally described and shared with others to benefit decision making in the domain and can help educate students and facilitate their progress toward expertise. The special status of the knowledge of experts in their domain of expertise is acknowledged even as far back as the Greek civilization. Socrates said that

> *I observe that when a decision has to be taken at the state assembly about some matter of building, they send for the builders to give their advice about the buildings, and when it concerns shipbuilding they send for the shipwrights, and similarly in every case where they are dealing with a subject which they think can be learned and taught. But if anyone else tries to give advice, whom they don't regard as an expert, no matter how handsome or*

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wealthy or well-born he is, they still will have none of him, but jeer at him and create an uproar, until either the would-be speaker is shouted down and gives up of his own accord, or else the police drag him away or put him out on the order of the presidents. (Plato, 1991, pp. 11–12)

Aristotle relied on his own senses as the primary source of scientific knowledge and sought out beekeepers, fishermen, hunters, and herdsmen to get the best and most reliable information for his books on science (Barnes, 2000). He even tried to explain occasional incorrect reports from some of his informants about how offspring of animals were generated. For example, some of them suggested that "the ravens and the ibises unite at the mouth" (Aristotle, 2000, p. 315). But Aristotle notes: "It is odd, however, that our friends do not reason out how the semen manages to pass through the stomach and arrive in the uterus, in view of the fact that the stomach concocts everything that gets into it, as it does the nourishment" (pp. 315 & 317). Similarly, "those who assert that the female fishes conceive as a result of swallowing the male's semen have failed to notice certain points" (p. 311). Aristotle explains that "Another point which helps to deceive these people is this. Fish of this sort take only a very short time over their copulation, with the result that many fishermen even never see it happening, for of course no fishermen ever watches this sort of thing for the sake of pure knowledge" (p. 313). Much of Aristotle's knowledge comes, at least partly, from consensus reports of professionals.

Much later during the Middle Ages, craftsmen formed guilds to protect themselves from competition. Through arrangements with the mayor and/or monarch they obtained a monopoly on providing particular types of handcraft and services with set quality standards (Epstein, 1991). They passed on their special knowledge of how to produce products, such as lace, barrels, and shoes, to their students (apprentices). Apprentices would typically start at around age 14 and commit to serve and study with their master for around 7 years – the length of time varied depending on the complexity of the craft and the age and prior experience of the apprentice (Epstein, 1991). Once an apprentice had served out their contract they were given a letter of recommendation and were free to work with other masters for pay, which often involved traveling to other cities and towns – they were therefore referred to as journeymen. When a journeyman had accumulated enough additional skill and saved enough money, he, or occasionally she, would often return to his home town to inherit or purchase a shop with tools and apply to become a master of the guild. In most guilds they required inspection of the journeyman's best work, that is, master pieces, and in some guilds they administered special tests to assess the level of performance (Epstein, 1991). When people were accepted as masters they were held responsible for the quality of the products from their shop and were thereby allowed to take on the training of apprentices (See Amirault & Branson, Chapter 5, and Chi, Chapter 2, on the progression toward expertise and mastery of a domain).

In a similar manner, the scholars' guild was established in the 12th and 13th century as "a *universitas magistribus et pupillorum*," or "guild of masters and students" (Krause, 1996, p. 9). Influenced by the University of Paris, most universities conducted all instruction in Latin, where the students were initially apprenticed as arts students until they successfully completed the preparatory (undergraduate) program and were admitted to the more advanced programs in medicine, law, or theology. To become a master, the advanced students needed to satisfy "a committee of examiners, then publicly defending a thesis, often in the town square and with local grocers and shoemakers asking questions" (Krause, 1996, p. 10). The goal of the universities was to accumulate and explain knowledge, and in the process masters organized the existing knowledge (See Amirault & Branson, Chapter 5). With the new organization of the existing knowledge of a domain, it was no longer necessary for individuals to discover the relevant knowledge and methods by themselves.

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Today's experts can rapidly acquire the knowledge originally discovered and accumulated by preceding expert practitioners by enrolling in courses taught by skilled and knowledgeable teachers using specially prepared textbooks. For example, in the 13th century Roger Bacon argued that it would be impossible to master mathematics by the then-known methods of learning (self-study) in less than 30 to 40 years (Singer, 1958). Today the roughly equivalent material (calculus) is taught in highly organized and accessible form in every high school.

Sir Francis Bacon is generally viewed as one of the architects of the Enlightenment period of Western Civilization and one of the main proponents of the benefits of generating new scientific knowledge. In 1620 he described in his book *Novum Organum* his proposal for collecting and organizing all existing knowledge to help our civilization engage in learning to develop a better world. In it, he appended a listing of all topics of knowledge to be included in *Catalogus Historarium Particularium*. It included a long list of skilled crafts, such as "History of weaving, and of ancillary skills associated with it," "History of dyeing," "History of leather-working, tanning, and of associated ancillary skills" (Rees & Wakely, 2004, p. 483).

The guilds guarded their knowledge and their monopoly of production. It is therefore not surprising that the same forces that eventually resulted in the French revolution were directed not only at the oppression by the king and the nobility, but also against the monopoly of services provided by the members of the guilds. Influenced by Sir Francis Bacon's call for an encyclopedic compilation of human knowledge, Diderot and D'Alembert worked on assembling all available knowledge in the first *Encyclopedie* (Diderot & D'Alembert, 1966–67), which was originally published in 1751–80.

Diderot was committed to the creation of comprehensive descriptions of the mechanical arts to make their knowledge available to the public and to encourage research and development in all stages of production and all types of skills, such as tannery, carpentry, glassmaking, and ironworking (Pannabecker, 1994), along with descriptions of how to sharpen a feather for writing with ink, as shown in Figure 1.1. His goal was to describe all the raw materials and tools that were necessary along with the methods of production. Diderot and his associate contributors had considerable difficulties gaining access to all the information because of the unwillingness of the guild members to answer their questions. Diderot even considered sending some of his assistants to become apprentices in the respective skills to gain access to all the relevant information (Pannabecker, 1994). In spite of all the information and pictures (diagrams of tools, workspaces, procedures, etc., as is illustrated in Figure 1.2 showing one of several plates of the process of printing) provided in the *Encyclopedie*, Diderot was under no illusion that the provided information would by itself allow anyone to become a craftsman in any of the described arts and wrote: "It is handicraft that makes the artist, and it is not in Books that one can learn to manipulate" (Pannabecker, 1994, p. 52). In fact, Diderot did not even address the higher levels of cognitive activity, "such as intuitive knowledge, experimentation, perceptual skills, problem-solving, or the analysis of conflicting or alternative technical approaches" (Pannabecker, 1994, p. 52).

A couple of years after the French revolution the monopoly of the guilds as eliminated (Fitzsimmons, 2003), including the restrictions on the practice of medicine and law. After the American Revolution and the creation of the United States of America laws were initially created to require that doctors and lawyers be highly trained based on the apprenticeship model, but pressure to eliminate elitist tendencies led to the repeal of those laws. From 1840 to the end of the 19th century there was no requirement for certification to practice medicine and law in the United States (Krause, 1996). However, with time both France and America realized the need to restrict vital medical and legal services to qualified professionals and developed procedures for training and certification.

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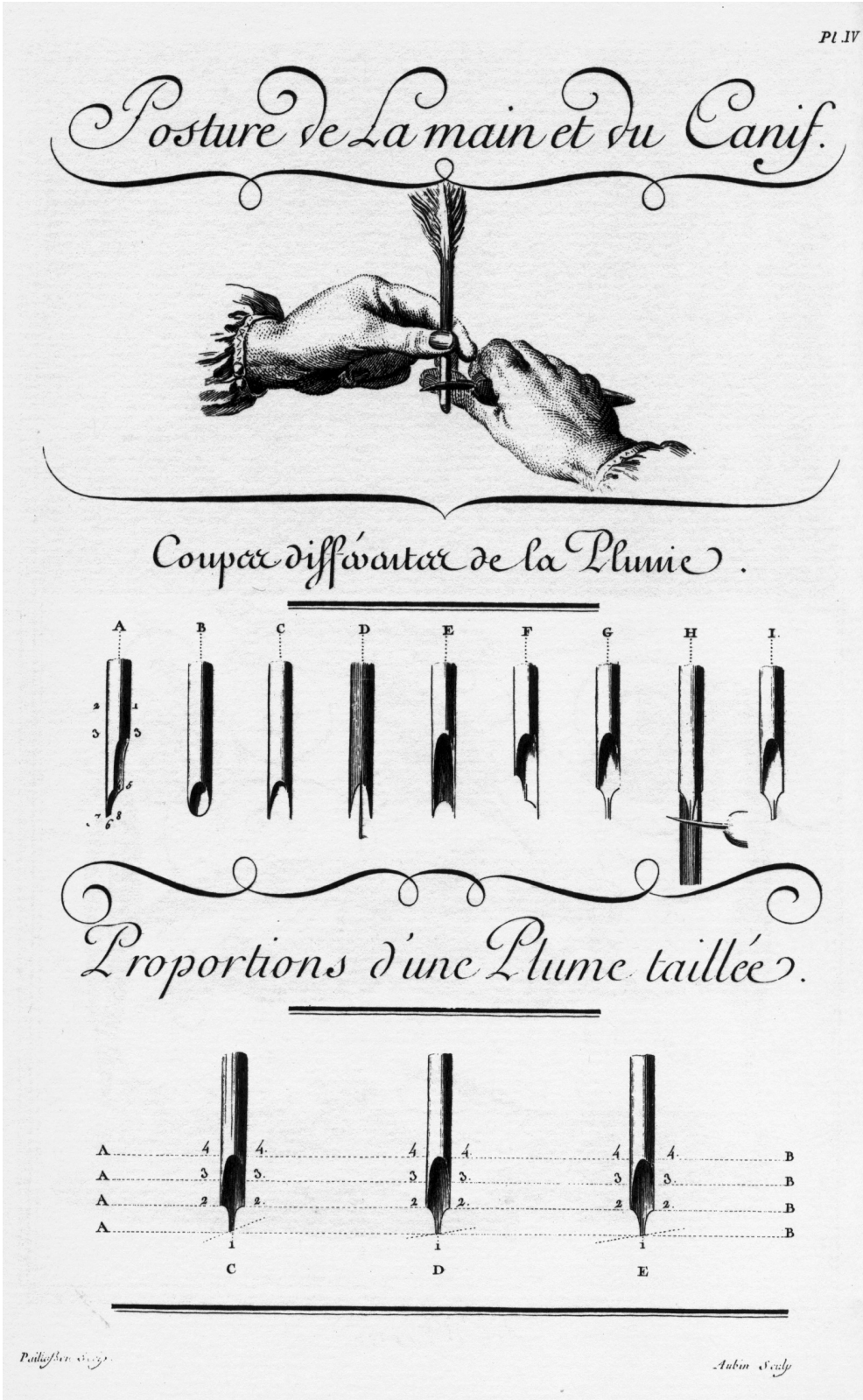


Figure 1.1. An illustration for how to sharpen a goose feather for writing with ink from Plate IV in the entry on "Ecriture" in the 23rd volume of *Encyclopedie ou dictionnare de raisonne des sciences, des artes et des métier* (Diderot & D'Alembert, 1966–67).

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Figure 1.2. An illustration of the workspace of a printer with some of his type elements from Plate I in the entry on "Imprimerie" in the 28th volume of *Encyclopedie ou dictionnare de raisonne des sciences, des artes et des métier* (Diderot & D'Alembert, 1966–67).

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Over the last couple of centuries there have been several major changes in the relation between master and apprentice. For example, before the middle of the 19th century children of poor families would often be taken on by teachers in exchange for a contractual claim for part of the future dancers', singers', or musicians' earnings as an adult (Rosselli, 1991). Since then the state has gotten more involved in the training of their expert performers, even outside the traditional areas of academia and professional training in medicine, law, business, and engineering. In the late 19th century, public institutions such as the Royal Academy of Music were established to promote the development of very high levels of skill in music to allow native students compete with better trained immigrants (Rohr, 2001). In a similar manner during the latter part of the 20th century, many countries invested in schools and academies for the development of highly skilled athletes for improved success in competitions during the Olympic Games and World Championships (Bloomfield, 2004).

More generally, over the last century there have been economic developments with public broadcasts of competitions and performances that generate sufficient revenue for a number of domains of expertise, such as sports and chess, to support professional full-time performers as well as coaches, trainers, and teachers. In these new domains, along with the traditional professions, current and past expert performers continue to be the primary teachers at the advanced level (masters), and their professional associations have the responsibility of certifying acceptable performance and the permission to practice. Accordingly, they hold the clout in thus influencing training in professional schools, such as law, medical, nursing, and business schools – "testing is the tail that wags the dog" (Feltovich, personal communication) – as well as continuing education training (see Evetts, Meig, & Felt, Chapter 7 on sociological perspectives on expertise). The accumulation of knowledge about the structure and acquisition of expertise in a given domain, as well as knowledge about the instruction and training of future professionals, has occurred, until quite recently, almost exclusively within each domain with little cross-fertilization of domains in terms of teaching, learning methods, and skill-training techniques.

It is not immediately apparent what is generalizable across such diverse domains of expertise, such as music, sport, medicine, and chess. What could possibly be shared by the skills of playing difficult pieces by Chopin, running a mile in less than four minutes, and playing chess at a high level? The premise for a field studying expertise and expert performance is that there are sufficient similarities in the theoretical principles mediating the phenomena and the methods for studying them in different domains that it would be possible to propose a general theory of expertise and expert performance. All of these domains of expertise have been created by humans. Thus the accumulated knowledge and skills are likely to reflect similarities in structure that reflect both human biological and psychological factors, as well as cultural factors. This raises many challenging problems for methodologies used to describe the organization of knowledge and mechanisms and reveals the mediating expert performance that generalizes across domains.

Once we know how experts organize their knowledge and their performance, is it possible to improve the efficiency of learning to reach higher levels of expert performance in these domains? It should also be possible to answer why different individuals improve their performance at different rates and why different people reach very different levels of final achievement. Would a deeper understanding of the development and its mediating mechanisms make it possible to select individuals with unusual potential and to design better developmental environments to increase the proportion of performers who reach the highest levels? Would it be possible even to facilitate the development of those rare individuals who make major creative contributions to their respective domains?

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Conceptions of Generalizable Aspects of Expertise

Several different theoretical frameworks have focused on broad issues on attaining expert performance that generalize across different domains of expertise.

Individual Differences in Mental Capacities

A widely accepted theoretical concept argues that general innate mental capacities mediate the attainment of exceptional performance in most domains of expertise. In his famous book, "*Heriditary Genius*," Galton (1869/1979) proposed that across a wide range of domains of intellectual activity the same innate factors were required to attain outstanding achievement and the designation of being a genius. He analyzed eminent individuals in many domains in Great Britain and found that these eminent individuals were very often the offspring of a small number of families – with much higher frequency than could be expected by chance. The descendents from these families were much more likely to make eminent contributions in very diverse domains of activity, such as becoming famous politicians, scientists, judges, musicians, painters, and authors. This observation led Galton to suggest that there must be a heritable potential that allows some people to reach an exceptional level in any one of many different domains. After reviewing the evidence that height and body size were heritable Galton (1869/1979) argued: "Now, if this be the case with stature, then it will be true as regards every other physical feature – as circumference of head, size of brain, weight of grey matter, number of brain fibres, &c.; and thence, a step on which no physiologist will hesitate, as regards mental capacity" (pp. 31–32, emphasis added).

Galton clearly acknowledged the need for training to reach high levels of performance in any domain. However, he argued that improvements are rapid only in the *beginning* of training and that subsequent increases become increasingly smaller, until "maximal performance becomes a rigidly determinate quantity" (p. 15). Galton developed a number of different mental tests of individual differences in mental capacity. Although he never related these measures to the objective performance of experts on particular real-world tasks, his views led to the common practice of using psychometric tests for admitting students into professional schools and academies for arts and sports with severely limited availability of slots. These tests of basic ability and talent were believed to identify the students with the capacity for reaching the highest levels.

In the 20th century scientists began the psychometric testing of large groups of experts to measure their powers of mental speed, memory, and intelligence. When the experts' performance was compared to control groups of comparable education, there was no evidence for Galton's hypothesis of a general superiority for experts because the demonstrated superiority of experts was found to be limited to specific aspects related to the particular domain of expertise. For example, the superiority of the chess experts' memory was constrained to regular chess positions and did not generalize to other types of materials (Djakow, Petrowski, & Rudik, 1927). Not even IQ could distinguish the best among chess players (Doll & Mayr, 1987) or the most successful and creative among artists and scientists (Taylor, 1975).

In a recent review, Ericsson and Lehmann (1996) found that (1) measures of basic mental capacities are not valid predictors of attainment of expert performance in a domain, (2) the superior performance of experts is often very domain specific, and transfer outside their narrow area of expertise is surprisingly limited, and (3) systematic differences between experts and less proficient individuals nearly always reflect attributes acquired by the experts during their lengthy training. The reader is directed to the chapter by Horn and Masunaga (chapter 34) and to comprehensive reviews in Sternberg and Grigorenko, 2003, and Howe, Davidson, and Sloboda. 1998.

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Expertise as the Extrapolation of Everyday Skill to Extended Experience

A second general type of theoretical frameworks is based on the assumption that the same learning mechanisms that account for the acquisition of everyday skills can be extended to the acquisition of higher levels of skills and expertise. Studies in the 19th century proposed that the acquisition of high levels of skills was a natural consequence of extended experience in the domains of expertise. For example, Bryan and Harter (1899) argued that ten years of experience were required to become a professional telegrapher. The most influential and pioneering work on expertise was conducted in the 1940s by Adrian de Groot (1978), who invited international chess masters and skilled club players to "think aloud" while they selected the best move for chess positions. His analyses of the protocols showed that the elite players were able to recognize and generate chess moves that were superior to skilled club players by relying on acquired patterns and planning (see Gobet & Charness, chapter 30, and Ericsson, chapter 13, for a more detailed account). DeGroot's dissertation was later translated into English in the late 1960s and early 1970s (deGroot, 1978) and had substantial impact on the seminal theory of expertise proposed by Herb Simon and Bill Chase (Simon & Chase, 1973).

In the 1950s and 1960s Newell and Simon proposed how information-processing models of human problem solving could be implemented as computer programs, such as the General Problem Solver (Ernst & Newell, 1969). In their seminal book, *Human Problem Solving*, Newell and Simon (1972) argued that domain-general problem solving was limited and that the thinking involved in solving most tasks could be represented as the execution of a sequence of production rules – such as IF <pattern>, THEN <action> – that incorporated specific knowledge about the task environment. In their theory of expertise, Simon and Chase (1973) made the fundamental assumption that the same patterns (chunks) that allowed the experts to retrieve suitable actions from memory were the same patterns that mediated experts' superior memory for the current situation in a game. Instead of studying the representative task of playing chess, namely, selecting the best moves for chess positions (Ericsson & Smith, 1991b; Vicente & Wang, 1998), Chase and Simon (1973) redirected the focus of research toward studying performance of memory tasks as a more direct method of studying the characteristics of patterns that mediate improvement in skill. They found that there was a clear relation between the number of chess pieces recalled from briefly presented chess positions and the player's level of chess expertise. Grand masters were able to reproduce almost the entire chessboards (24 to 26 pieces) by recalling a small number of complex chunks, whereas novices could recall only around 4 pieces, where each piece was a chunk. The masters' superior memory was assumed to depend on an acquired body of many different patterns in memory because their memory for randomly rearranged chess configurations was markedly reduced. In fact in such configurations they could recall only around 5 to 7 pieces, which was only slightly better than the recall of novices.

Experts' superiority for representative but not randomly rearranged stimuli has since been demonstrated in a large number of domains. The relation between the mechanisms mediating memory performance and the mechanisms mediating representative performance in the same domains have been found to be much more complex than originally proposed by Simon and Chase (1973) (see Gobet & Charness, Chapter 30, and Wilding & Valentine, Chapter 31. See also Ericsson & Kintsch, 1995; Ericsson, Patel, & Kintsch, 2000; Gobet & Simon, 1996; Simon & Gobet, 2000; Vicente & Wang, 1998).

Expertise as Qualitatively Different Representation and Organization of Knowledge

A different family of approaches drawing on the Simon-Chase theory of expertise has focused on the content and organization of

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the experts' knowledge (Chi, Feltovich, & Glaser, 1981; Chi, Glaser, & Rees, 1982) and on methods to extract the experts knowledge to build computer-based models emulating the experts' performance (Hoffman, 1992). These approaches have studied experts, namely, individuals who are socially recognized as experts and/or distinguished by their extensive experience (typically over 10 years) and by knowledge of a particular subject attained through instruction, study, or practical experience. The work of Robert Glaser, Micheline Chi, and Paul Feltovich examined the representations of knowledge and problem solutions in academic domains, such as physics (See Chi, Chapters 3 and 10). Of particular importance, Chi studied children with extensive knowledge of chess and dinosaurs (See Chi, Chapter 10), and found these children displayed many of the same characteristics of the knowledge representation of adult experts. This work on expertise is summarized in Feltovich, Prietula, and Ericsson, Chapter 4, Chi, Chapter 10, and Hoffman and Lintern, Chapter 12, and in a couple of edited volumes (Chi, Glaser, & Farr, 1988; Starkes & Allard, 1993).

In a parallel development in the computer science of the late 1970s and early 1980s, Ed Feigenbaum and other researchers in the area of artificial intelligence and cognitive science have attempted to elicit the knowledge of experts (Hoffman, 1992) and to incorporate their knowledge in computer models (c.f. expert systems) that seek to replicate some of the decision making and behavior of experts (see Buchanan, Davis, & Feigenbaum, Chapter 6, and Hoffman & Lintern, Chapter 12). There has been a longstanding controversy over whether highly experienced experts are capable of articulating the knowledge and methods that control their generation of appropriate actions in complex situations.

The tradition of skill acquisition of Bryan and Harter (1899), Fitts and Posner (1967), and Simon and Chase (1973) assumed that expert performance was associated with automation and was virtually effortless performance based on pattern recognition and direct access of actions. However, Polanyi (1962, 1966) is generally recognized as the first critic who saw that nonconscious and intuitive mediation limits the possibility of eliciting and mapping the knowledge and rules that mediates experts' intuitive actions. Subsequent discussion of the development of expertise by Dreyfus and Dreyfus (1986) and Benner (1984) has argued that the highest levels of expertise are characterized by contextually based intuitive actions that are difficult or impossible to report verbally. Several chapters in this handbook propose methods for uncovering tacit knowledge about the successful development of expertise (Cianciolo, Matthew, Wagner, & Sternberg, Chapter 35), about methods of work through observation (Clancey, Chapter 8), Concept Mapping (Hoffman & Lintern, Chapter 12), similarity judgment (Chi, Chapter 10), and traditional psychometric analyses of individual differences in performance (Ackerman & Beier, Chapter 9) or simulated environments (Ward, Williams, & Hancock, Chapter 14). Other investigators argue that expert performers often continue to engage in deliberate practice in order to improve and that these performers have to actively retain and refine their mental representations for monitoring and controlling their performance. This retained ability to monitor performance allows them to give informative concurrent and retrospective reports about the mediating sequences of thoughts (see Ericsson, Chapter 13).

Expertise as Elite Achievement Resulting from Superior Learning Environments

There are other approaches to the study of expertise that have focused on objective achievement. There is a long tradition of influential studies with interviews of peer-nominated eminent scientists (Roe, 1952) and analyses of biographical data on Nobel Prize winners (Zuckerman, 1977) (see Simonton, Chapter 18, 1994, for a more extensive account). In a seminal study, Benjamin Bloom and his colleagues (Bloom, 1985a) interviewed international-level performers from six different domains of expertise ranging from swimming to molecular

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genetics. All of the 120 participants had won prizes at international competitions in their respective domains. They were all interviewed about their development, as were their parents, teachers, and coaches. For example, Bloom and his colleagues collected information on the development of athletes who had won international competitions in swimming and tennis. They also interviewed artists who have won international competitions in sculpting and piano playing and scientists who had won international awards in mathematics and molecular biology. In each of these six domains Bloom (1985b) found evidence for uniformly favorable learning environments for the participants. Bloom (1985b) concluded that the availability of early instruction and support by their family appeared to be necessary for attaining an international level of performance as an adult. He found that the elite performers typically started early to engage in relevant training activities in the domain and were supported both by exceptional teachers and committed parents. One of the contributors to the *Handbook*, Lauren Sosniak (1985a, 1985b, 1985c, 1985d), describes in Chapter 16 the main findings from the original study (Bloom, 1985a), along with more recent interview studies aimed to uncover the development of elite performers.

Expertise as Reliably Superior (Expert) Performance on Representative Tasks

It is difficult to identify the many mediating factors that might have been responsible for the elite performer to win an award and to write a groundbreaking book. When eminence and expertise is based on a singular or small number of unique creative products, such as books, paintings, or musical as compositions, it is rarely possible to identify and study scientifically the key factors that allowed these people to produce these achievements. Consequently, Ericsson and Smith (1991b) proposed that the study of expertise with laboratory rigor requires representative tasks that capture the essence of expert performance in a specific domain of expertise. For example, a world-class sprinter will be able to reproduce superior running performance on many tracks and even indoors in a large laboratory. Similarly, de Groot (1978) found that the ability to select the best move for presented chess positions is the best correlate of chess ratings and performance at chess tournaments – a finding that has been frequently replicated (Ericsson & Lehmann, 1996; van der Maas & Wagenmakers, 2005). Once it is possible to reproduce the reliably superior performance of experts in a controlled setting, such as a laboratory, it then becomes feasible to examine the specific mediating mechanisms with experiments and process-tracing techniques, such as think aloud verbal reports (see Ericsson, Chapter 13, and Ericsson & Smith, 1991b). The discovery of representative tasks that measure adult expert performance under standardized conditions in a controlled setting, such as a laboratory, makes it possible to measure and compare the performance of less-skilled individuals on the same tasks. Even more important, it allows scientists to test aspiring performers many times during their development of expertise, allowing the measurement of gradual increases in performance.

The new focus on the measurement of expert performance with standardized tasks revealed that "experts," that is, individuals identified by their reputation or their extensive experience, are not always able to exhibit reliably superior performance. There are at least some domains where "experts" perform no better than less-trained individuals and that sometimes experts' decisions are no more accurate than beginners' decisions and simple decision aids (Camerer & Johnson, 1991; Bolger & Wright, 1992). Most individuals who start as active professionals or as beginners in a domain change their behavior and increase their performance for a limited time until they reach an acceptable level. Beyond this point, however, further improvements appear to be unpredictable and the number of years of work and leisure experience in a domain is a poor predictor of attained performance (Ericsson & Lehmann, 1996). Hence, continued improvements (changes)

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in achievement are not automatic consequences of more experience, and in those domains where performance consistently increases, aspiring experts seek out particular kinds of experience, that is, deliberate practice (Ericsson, Krampe, & Tesch-Römer, 1993). Such activities are designed, typically by a teacher, for the sole purpose of effectively improving specific aspects of an individual's performance. A large body of research shows how deliberate practice can change mediating mechanisms and that the accumulated amounts of deliberate practice are related to the attained level of performance (see Ericsson, Chapter 38, and Deakin, Coté, & Harvey, Chapter 17, Zimmerman, Chapter 38, as well as the edited books by Ericsson [1996a] and Starkes & Ericsson [2003]).

General Comments

In summary, there are a broad range of approaches to the study of the structure and acquisition of expertise as well as expert performance. Although individual researchers and editors may be committed to one approach over the others, this *Handbook* has been designed to fairly cover a wide range of approaches and research topics in order to allow authors to express their different views. However, the authors have been encouraged to describe explicitly their empirical criteria for their key terms, such as "experts" and "expert performance." For example, the authors have been asked to report if the cited research findings involve experts identified by social criteria, criteria of lengthy domain-related experience, or criteria based on reproducibly superior performance on a particular set of tasks representative of the individuals' domain of expertise.

General Outline of the Handbook

The handbook is organized into six general sections. First, Section 1 introduces the *Handbook* with brief accounts of general perspectives on expertise. In addition to this introductory chapter that outlines the organization of the handbook, there are chapters by two of the pioneers of the study of cognitive skill and expertise. Michelene Chi (Chapter 2) describes two approaches to the study of expertise and Earl Hunt (Chapter 3) gives his general perspective on the principal factors related to expertise. In a recent book Hunt (1995) has made a convincing case for the increasing importance of high levels of skill in occupations of the future. He argues that with the development of technology to automate less complex jobs the most important occupations of the future will require creative design and planning that cannot be easily automated. He foresees a rapidly increasing need to train students to even higher levels of expertise to continue the development of our modern society. The key competitive differences between companies of the future may not have to do with raw materials and monetary resources but with human capital, namely, the abilities of the employees. The Nobel Prize winner Gary Becker has for a long time made the case for the critical role of education and human capital in our current industrialized world, and especially the crucial role of highly accomplished people. He (Becker, 2002) illustrated this claim by a quote from Microsoft founder Bill Gates: Take our 20 best people away and . . . Microsoft would become an unimportant company" (Becker, 2002, p. 8).

The second section of the *Handbook* contains reviews of the historical development of the study of expertise in four major disciplines, namely, psychology, education, computer science, and sociology. Three pioneers in the psychological study of expertise, Paul Feltovich, Michael Prietula, and Anders Ericsson, describe the development of the study of expertise in psychology (Chapter 4). One of the pioneers in the development of instructional design, Robert Branson, has together with Ray Amirault (Chapter 5) described the role of expertise in the historical development of educational methods and theories. Three of the pioneers in the development of expert systems, Bruce Buchanan, Randall Davis, and Edward Feigenbaum (Chapter 6), describe the role of expertise in shaping contemporary approaches in computer science and

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artificial intelligence. Finally, Julia Evetts, Harald Mieg, and Ulrike Felt (Chapter 7) provide a description of the relevant approaches to the study of expertise from the point of view of sociology.

The next two sections of the *Handbook* review the core methods for studying the structure (Section 3) and acquisition (Section 4) of expertise and expert performance. Each of the chapters in Sections 3 and 4 has been written by one of the pioneering researchers who have developed these methods and approaches for use in research on expertise and expert performance. The chapters consist of a historical background, a detailed description of the recommended methodology with a couple of examples, and a general review of the type of empirical evidence that has been collected. In the first chapter of Section 3 William Clancey (Chapter 8) gives an overview of the ethnographic observational methods for studying the behavior of experts. Philip Ackerman and Margaret Beier (Chapter 9) review the use of psychometric methods for studying expertise. Michelene Chi (Chapter 10) describes how laboratory methods have been used to assess the structure of knowledge. Jan Maarten Schraagen (Chapter 11) describes how tasks presented to skilled and less-skilled individuals can be analyzed and how a task analysis can guide data analysis and theory construction. Robert Hoffman and Gavin Lintern (Chapter 12) review methods for how knowledge of experts can be elicited and represented by interviews, Concept Maps, and abstraction-decomposition diagrams. Anders Ericsson (Chapter 13) describes how the elicitation of "think-aloud" protocols can allow investigators to trace the thought processes of experts while they perform representative tasks from their domain. Finally, Paul Ward, Mark Williams, and Peter Hancock (Chapter 14) review how simulated environments can both be used to measure experts' representative performance as well as be used for training.

Section 4 contains chapters examining methods for studying how skill, expertise, and expert performance develop and are acquired through practice. In the first chapter, Robert Proctor and Kim-Phuon Vu (Chapter 15) describe how laboratory methods for the study of skilled performance can inform research on expertise and expert performance. Lauren Sosniak (Chapter 16) discusses how she and her colleagues used retrospective interviews to describe the development of expertise in the classic studies led by Benjamin Bloom (1985a), along with some recent extensions of that work. Janice Deakin, Jean Côté, and Andrew Harvey (Chapter 17) use diaries and describe different methods to study how expert performers spend their time and how experts allocate their practice time. In the final chapter of this section, Dean Simonton (Chapter 18) reviews the methods of historiometrics and how data about the development of eminent performers can be collected and analyzed.

Section 5 consists of fifteen chapters that review our current knowledge about expertise and expert performance in particular domains and represents the core of this *Handbook*. Each chapter has been written by internationally respected experts on the associated areas of expertise and contains a brief historic background followed by a review and future directions. The chapters in Section 5 have been broken down into three subsections. The first subsection is focused on different types of professional expertise, namely, medicine (Chapter 19 by Geoff Norman, Kevin Eva, Lee Brooks, and Stan Hamstra), transportation, such as driving, flying, and airplane control (Chapter 20 by Francis Durso and Andrew Dattel), software design (Chapter 21 by Sabine Sonnentag, Cornelia Niessen, and Judith Volmer), and writing (Chapter 22 by Ronald Kellogg). There are two chapters on various aspects of decision making, namely, judgments in dynamic situations (natural decision making, Chapter 23 by Karol Ross, Jennifer Shafer, and Gary Klein) and decision-making expertise (Chapter 24 by Frank Yates & Michael Tschirhart), followed by Chapter 25 by Eduardo Salas, Michael Rosen, Shawn Burke, Gerald Goodwin, and Stephen Fiore on research on expert teams. The second subsection contains chapters that review expert performance in music

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(Chapter 26 by Andreas Lehmann and Hans Gruber) and in sports (Chapter 27 by Nicola Hodges, Janet Starkes, and Clare MacMahon), and expertise in other types of arts, such as acting, ballet, and dance (Chapter 28 by Helga Noice and Tony Noice). The final chapter in this subsection reviews research on perceptual-motor skills (Chapter 29 by David Rosenbaum, Jason Augustyn, Rajal Cohen, and Steven Jax). The third and final subsection covers the findings in a diverse set of domains of expertise, including games. The first chapter (Chapter 30 by Fernand Gobet and Neil Charness) describes the pioneering and influential work on expertise in the game of chess. The next chapter (Chapter 31 by John Wilding and Elizabeth Valentine) reviews research on exceptional memory, in particular for information that most people have difficulty remembering, such as numbers, names, and faces. The last two chapters review research on mathematical ability and expertise (Chapter 32 by Brian Butterworth) and expertise in history (Chapter 33 by Jim Voss and Jennifer Wiley) – an example of a knowledge-based domain.

In the last section of the *Handbook* we have invited some of the world's leading researchers on general theoretical issues that are cutting across different domains of expertise to review the current state of knowledge. In the first chapter John Horn and Hiromi Masunaga (Chapter 34) discuss the relation between general intelligence and expertise. In the following chapter Anna Cianciolo, Cynthia Mattew, Richard Wagner, and Robert Sternberg (Chapter 35) review the relation between expertise and central concepts, such as practical intelligence and tacit knowledge. Mica Endsley (Chapter 36) reviews evidence for situational awareness, namely, experts' superior ability to perceive and monitor critical aspects of situations during performance. The next three chapters focus on aspects of learning. Nicole Hill and Walter Schneider (Chapter 37) review the neurological evidence on physiological adaptations resulting from the acquisition of expertise. Anders Ericsson (Chapter 38) reviews the evidence for the key role of deliberate practice in causing physiological adaptations and the acquisition of mechanisms that mediate expert performance. Finally, Barry Zimmerman (Chapter 39) describes the importance of self-regulated learning in the development of expertise. The last three chapters review general issues in expertise. Ralf Krampe and Neil Charness (Chapter 40) review the effects of aging on expert performance and how it might be counteracted. Harald Mieg (Chapter 41) reviews the importance of social factors in the development of expertise. Finally, Robert Weisberg (Chapter 42) discusses the relation between expertise and creativity.

Conclusion

This *Handbook* has been designed to provide researchers, students, teachers, coaches, and anyone interested in attaining expertise with a comprehensive reference to methods, findings, and theories related to expertise and expert performance. It can be an essential tool for researchers, professionals, and students involved in the study or the training of expert performance and a necessary source for college and university libraries, as well as public libraries. In addition, the *Handbook* is designed to provide a suitable text for graduate courses on expertise and expert performance. More generally, it is likely that professionals, graduate students, and even undergraduates who aspire to higher levels of performance in a given domain can learn from experts' pathways to superior performance in similar domains.

Many researchers studying expertise and expert performance are excited and personally curious about the established research findings that most types of expertise require at least a decade of extended efforts to attain the mechanisms mediating superior performance. There is considerable knowledge that is accumulating about generalizations across many domains about the acquisition and refinement of these mechanisms during an extended period of deliberate practice. The generalizable insights range from the

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characteristics of ideal training environments, to the methods for fostering motivation by providing both emotional support and attainable training tasks of a suitable difficulty level. This theoretical framework has several implications.

It implies that if someone is interested in the upper limits of human performance and the most effective training to achieve the highest attainable levels, they should study the training techniques and performance limits of experts who have spent their entire life maximizing their performance. This assumption also implies that the study of expert performance will provide us with the best current evidence on what is humanly possible to achieve with today's methods of training and how these elite performers are able to achieve their highest levels of performance. Given that performance levels are increasing every decade in most domains of expertise, scientists will need to work with elite performers and their coaches to discover jointly the ever-increasing levels of improved performance.

The framework has implications for education and professional training of performance for all the preliminary levels that lead up to the expert levels in professional domains of expertise. By examining how the prospective expert performers attained lower levels of achievement, we should be able to develop practice environments and foster learning methods that help people to attain the fundamental representations of the tasks and the self-regulatory skills that were necessary for the prospective experts to advance their learning to higher levels.

With the rapid changes in the relevant knowledge and techniques required for most jobs, nearly everyone will have to continue their learning and even intermittently relearn aspects of their professional skills. The life-long quest for improved adaptation to task demands will not be limited to experts anymore. We will all need to adopt the characteristics and the methods of the expert performers who continuously strive to attain and maintain their best level of achievement.

References

Anderson, J. R. (Ed.) (1981). *Cognitive skills and their acquisition*. Hillsdale, NJ: Erlbaum .

Aristotle (2000). *Generation of animals* (Translated by A. L. Pick and first published in 1942). Cambridge, MA: Harvard Univesity Press.

Barnes, J. (2000). *Aristotle: A very short introduction*. Oxford: Oxford University Press.

Becker, G. S. (2002). The age of human capital. In E. P. Lazear (Ed.), *Education in the twenty-first century* (pp. 3–8). Stanford,CA: Hoover Institution Press.

Bédard, J. (1991). Expertise and its relation to audit decision quality. *Contemporary Accounting Research*, 8, 198–222.

Benner, P. E. (1984). *From novice to expert – Excellence and power in clinical nursing practice*. Menlo Park, CA: Addison-Wesley.

Bloom, B. S. (Ed.) (1985a). *Developing talent in young people*. New York: Ballantine Books.

Bloom, B. S. (1985b). Generalizations about talent development. In B. S. Bloom (Ed.), *Developing talent in young people* (pp. 507–549). New York: Ballantine Books.

Bloomfield, J. (2004). *Australia's sporting success: The inside story*. Sydney, Australia: University of South Wales Press.

Bolger, F., & Wright, G. (1992). Reliability and validity in expert judgment. In G. Wright and F. Bolger (Eds.), *Expertise and decision support* (pp. 47–76). New York: Plenum.

Bryan, W. L., & Harter, N. (1899). Studies on the telegraphic language: The acquisition of a hierarchy of habits. *Psychological Review*, 6, 345–375.

Camerer, C. F., & Johnson, E. J. (1991). The process-performance paradox in expert judgment: How can the experts know so much and predict so badly? In K. A. Ericsson and J. Smith (Eds.), *Towards a general theory of expertise: Prospects and limits* (pp. 195–217). Cambridge: Cambridge University Press.

Chase, W. G. (Ed.) (1973). *Visual information processing*. New York: Academic Press.

Chase, W. G., & Simon, H. A. (1973). The mind's eye in chess. In W. G. Chase (Ed.), *Visual information processing* (pp. 215–281). New York: Academic Press.

Chi, M. T. H., Feltovich, P. J., & Glaser, R. (1981). Categorization and representation of physics problems by experts and novices. *Cognitive Science*, 5, 121–152.

Downloaded from https://www.cambridge.org/core. University College London (UCL), on 02 May 2018 at 21:26:29, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9780511816796.001

Chi, M. T. H., Glaser, R., & Farr, M. J. (Eds.) (1988). *The nature of expertise*. Hillsdale, NJ: Erlbaum.

Chi, M. T. H., Glaser, R., & Rees, E. (1982). Expertise in problem solving. In R. S. Sternberg (Ed.). *Advances in the psychology of human intelligence* (Vol. 1, pp. 1–75). Hillsdale, NJ: Erlbaum.

De Groot, A. (1978). *Thought and choice in chess*. (Original by published in 1946). The Hague, the Netherlands: Mouton.

Diderot, D., & D'Alembert, J. L. R. (Eds.) (1966–67). *Encyclopedie ou dictionnaie de raisonne' des sciences, des artes et des métier [Encyclopedia or dictionary of the sciences, arts, and occupations]*. (Originally published 1751–80). Stuttgart-Bad Cannstatt, Gemany: Frommann.

Djakow, Petrowski, & Rudik (1927). *Psychologie des Schachspiels [Psychology of chess]*. Berlin: Walter de Gruyter.

Doll, J., & Mayr, U. (1987). Intelligenz und Schachleistung – eine Untersuchung an Schachexperten. [Intelligence and achievement in chess – a study of chess masters]. *Psychologische Beiträge*, 29, 270–289.

Dreyfus, H. L., & Dreyfus, S. E. (1986). *Mind over machine: The power of intuition and expertise in the era of the computer*. New York: The Free Press.

Epstein, S. A. (1991). *Wage, labor, & guilds in medieval Europe*. Chapel Hill, NC: North Carolina University Press.

Ericsson, K. A. (Ed.) (1996a). *The road to excellence: The acquisition of expert performance in the arts and sciences, sports, and games*. Mahwah, NJ: Erlbaum.

Ericsson, K. A. (1996b). The acquisition of expert performance: An introduction to some of the issues. In K. A. Ericsson (Ed.), *The road to excellence: The acquisition of expert performance in the arts and sciences, sports, and games* (pp. 1–50). Mahwah, NJ: Erlbaum.

Ericsson, K. A. (2004). Deliberate practice and the acquisition and maintenance of expert performance in medicine and related domains. *Academic Medicine*, 10, S70–S81.

Ericsson, K. A., & Kintsch, W. (1995). Long-term working memory. *Psychological Review*, 102, 211–245.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100, 363–406.

Ericsson, K. A., & Lehmann, A. C. (1996). Expert and exceptional performance: Evidence on maximal adaptations on task constraints. *Annual Review of Psychology*, 47, 273–305.

Ericsson, K. A., Patel, V. L., & Kintsch, W. (2000). How experts' adaptations to representative task demands account for the expertise effect in memory recall: Comment on Vicente and Wang (1998). *Psychological Review*, 107, 578–592.

Ericsson, K. A., & Smith, J. (Eds.) (1991a). *Toward a general theory of expertise: Prospects and limits*. Cambridge: Cambridge University Press.

Ericsson, K. A., & Smith, J. (1991b). Prospects and limits in the empirical study of expertise: An introduction. In K. A. Ericsson and J. Smith (Eds.), *Toward a general theory of expertise: Prospects and limits* (pp. 1–38). Cambridge: Cambridge University Press.

Ernst, G. W., & Newell, A. (1969). *GPS: A case study in generality and problem solving*. New York: Academic Press.

Feltovich, P. J., Ford, K. M., & Hoffman, R. R. (Eds.) (1997). *Expertise in context: Human and machine*. Cambridge, MA: AAAI/MIT Press.

Fitts, P., & Posner, M. I. (1967). *Human performance*. Belmont, CA: Brooks/Cole.

Fitzsimmons, M. P. (2003). *The night that the Old Regime ended*. University Park, PA: Pennsylvania State University Press.

Galton, F., Sir (1869/1979). *Hereditary genius: an inquiry into its laws and consequences*. (Originally published in 1869). London: Julian Friedman Publishers.

Gobet, F., & Simon, H. A. (1996). Templates in chess memory: A mechanism for recalling several boards. *Cognitive Psychology*, 31, 1–40.

Hoffman, R. R. (Ed.) (1992). *The psychology of expertise: Cognitive research and empirical AI*. New York: Springer-Verlag.

Howe, M. J. A., Davidson, J. W., & Sloboda, J. A. (1998). Innate talents: Reality or myth? *Behavioral and Brain Sciences*, 21, 399–442.

Hunt, E. B. (1995). *Will we be smart enough?* New York: Russell Sage Foundation.

Krause, E. A. (1996). *Death of guilds: Professions, states and the advance of capitalism, 1930 to the present*. New Haven, CT: Yale University Press.

Lyons, R., Payne, C., McCabe, M., & Fielder, C. (1998). Legibility of doctors' handwriting: Quantitative and comparative study. *British Medical Journal*, 317, 863–864.

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Newell, A., & Simon, H. A. (1972). *Human problem solving*. Englewood Cliffs, NJ: Prentice-Hall.

Pannabecker, J. R. (1994). Diderot, the mechanical arts and the *encyclopedie*: In search of the heritage of technology education. *Journal of Technology Education*, *6*, 45–57.

Plato (1991). *Protagoras* (Translated by C. C. W. Taylor). Oxford: Clarendon Press.

Polanyi, M. (1962). *Personal knowledge: Toward a post-critical philosophy* (Corrected edition). Chicago: The University of Chicago Press.

Polanyi, M. (1966). *The tacit dimension* (Reprinted in 1983). Gloucester, MA: Peter Smith.

Rees, G., & Wakely, M. (Eds.) (2004). *The instauratio magna, Part II: Novum organum and associated texts*. Oxford: Clarendon Press.

Roe, A. (1952). *The making of a scientist*. New York: Dodd, Mead & Company.

Rohr, D. (2001). *The careers and social status of British musicians, 1750–1850: A profession of artisans*. Cambridge: Cambridge University Press.

Rosselli, J. (1991). *Music & musicians in nineteenth-century Italy*. Portland, OR: Amadeus Press.

Shanteau, J. (1988). Psychological characteristics and strategies of expert decision makers. *Acta Psychologica*, *68*, 203–215.

Simon, H. A., & Gobet, F. (2000). Expertise effects in memory recall: Comment on Vicente and Wang (1998). *Psychological Review*, *107*, 593–600.

Simon, H. A, & Chase, W. G. (1973). Skill in chess. *American Scientist*, *61*, 394–403.

Simonton, D. K. (1994). *Greatness: Who makes history and why*. New York: Guilford Press.

Singer, C. (1958). *From magic to science*. New York: Dover.

Sosniak, L. A. (1985a). Learning to be a concert pianist. In B. S. Bloom (Ed.), *Developing talent in young people* (pp. 19–67). New York: Ballantine Books.

Sosniak, L. A. (1985b). Becoming an outstanding research neurologist. In B. S. Bloom (Ed.), *Developing talent in young people* (pp. 348–408). New York: Ballantine Books.

Sosniak, L. A. (1985c). Phases of learning. In B. S. Bloom (Ed.), *Developing talent in young people* (pp. 409–438). New York: Ballantine Books.

Sosniak, L. A. (1985d). A long-term commitment to learning. In B. S. Bloom (Ed.), *Developing talent in young people* (pp. 477–506). New York: Ballantine Books.

Starkes, J. L., & Allard, F. (Eds.) (1993). *Cognitive issues in motor expertise*. Amsterdam: North Holland.

Starkes, J., & Ericsson, K. A. (Eds.) (2003). *Expert performance in sport: Recent advances in research on sport expertise*. Champaign, IL: Human Kinetics.

Sternberg, R. J., & Grigorenko, E. L. (Eds.) (2003). *Perspectives on the psychology of abilities, competencies, and expertise*. Cambridge: Cambridge University Press.

Taylor, I. A. (1975). A retrospective view of creativity investigation. In I. A. Taylor and J. W. Getzels (Eds.), *Perspectives in creativity* (pp. 1–36). Chicago, IL; Aldine Publishing Co.

Van der Maas, H. L. J., & Wagenmakers, E. J. (2005). A psychometric analysis of chess expertise. *American Journal of Psychology*, *118*, 29–60.

Vicente, K. J., & Wang, J. H. (1998). An ecological theory of expertise effects in memory recall. *Psychological Review*, *105*, 33–57.

Webster's New World Dictionary (1968). Cleveland, OH: The World Publishing Company.

Wikipedia. http://en.wikipedia.org/wiki/Expert

Zuckerman, H. (1977). *Scientific elite: Nobel laureates in the United States*. New York: Free Press.

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